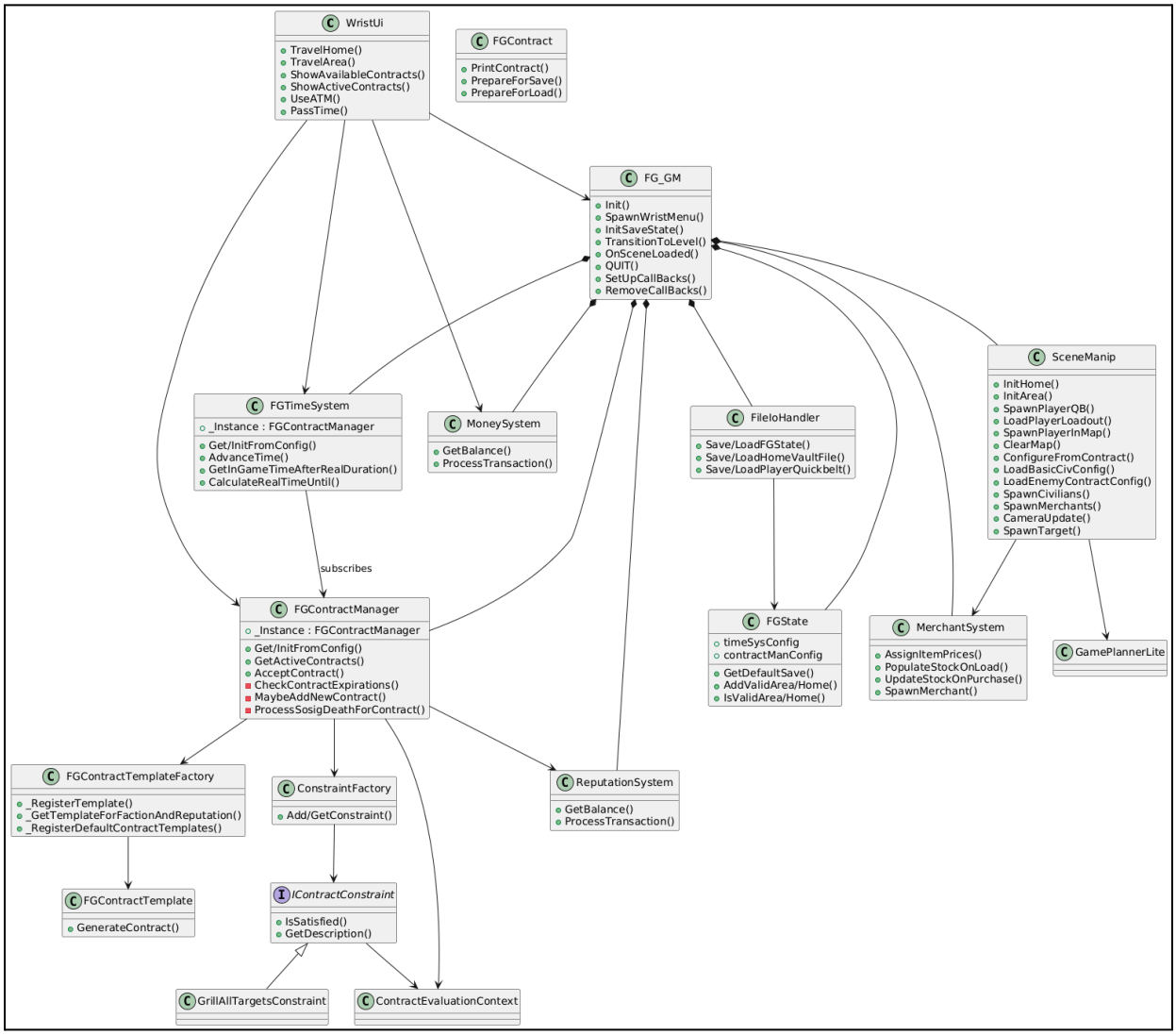
FG Components Design

Status: In progress

Naming convention will use “FreeGriller” and “FG”.



[Wrist-Tablet]

- Displays available and active contracts.
- Allows travel to different maps.
- Shows money balance and provides ATM access.
- Allows time progression.

[Maps-Loader] (static)

- Saves the player's current loadout before loading a map.
- Spawns the player in the selected map.
- Clears previous map state by default.
- Loads the player's saved loadout.
- Loads a base civilian scene configuration if available.
- Loads enemy contract configurations if a contract is active.
- Loads the player's safehouse if applicable.

[Merchant-System]

- Assigns item prices dynamically.
- Populates merchant stock when loading a map.
- Retains previously sold items (does not repopulate).
- Restocks after a set time interval.
- Manages stock separately for each merchant.

[Money-System]

- Tracks the player's bank account balance after purchases and earnings.

[Time-Keeper]

- Manages in-game time progression with a multiplier.
- Allows fast-forwarding time.
- Converts in-game time to a real-world equivalent.

[Contract-System] (static)

- Generates available and active contracts, replacing expired ones.
- Selects valid maps and target spawn locations based on scene configurations.
- Creates contract configurations with tier-based modifications.
- Loads contract configurations via **Contract-Loader** on map load.
- Tracks active contracts and linked Sosigs in the game world.
- Sets up payout modifications based on contract restrictions.
- Updates the player's money upon contract completion.

[Contract-Config]

- Defines contract-specific settings, including:
    - Map, civilian scene configuration, and enemy configuration.
    - Targets, guards, and additional entities with assigned **SosigEnemyId** keys.
    - Contract restrictions and base payout.
- Provides a function that executes upon contract completion.

[Contract-Restriction]

- Defines rules and conditions for contract completion.
- Attaches callback functions to Sosigs.
- Adjusts contract payouts based on in-mission events.

[Contract-Loader]

- Reads **Contract-Config** data.
- Finds the designated spawn points for targets and enemies.
- Assigns **SosigEnemyIds** to specific spawn locations.
- Spawns Sosigs, tracks them, and links them to the **Contract-System**.

```
@startuml

class WristUi {
    + TravelHome()
    + TravelArea()
    + ShowAvailableContracts()
    + ShowActiveContracts()
    + UseATM()
    + PassTime()
}

class SceneManip {
    + InitHome()
    + InitArea()
    + SpawnPlayerQB()
    + LoadPlayerLoadout()
    + SpawnPlayerInMap()
    + ClearMap()
    + ConfigureFromContract()
    + LoadBasicCivConfig()
    + LoadEnemyContractConfig()
    + SpawnCivilians()
    + SpawnMerchants()
    + CameraUpdate()
    + SpawnTarget()
}

class MerchantSystem {
    + AssignItemPrices()
```

```
    + PopulateStockOnLoad()
    + UpdateStockOnPurchase()
    + SpawnMerchant()
}

class MoneySystem {
    + GetBalance()
    + ProcessTransaction()
}

class ReputationSystem {
    + GetBalance()
    + ProcessTransaction()
}

class FGTimeSystem {
    + _Instance : FGContractManager
    + Get/InitFromConfig()
    + AdvanceTime()
    + GetInGameTimeAfterRealDuration()
    + CalculateRealTimeUntil()
}

class FGContractManager {
    + _Instance : FGContractManager
    + Get/InitFromConfig()
    + GetActiveContracts()
    + AcceptContract()
    - CheckContractExpirations()
    - MaybeAddNewContract()
    - ProcessSosigDeathForContract()
}

class FGContract {
    + PrintContract()
    + PrepareForSave()
    + PrepareForLoad()
}

class FGContractTemplateFactory {
    + _RegisterTemplate()
    + _GetTemplateForFactionAndReputation()
    + _RegisterDefaultContractTemplates()
```

```
}

class FGContractTemplate {
    + GenerateContract()
}

class ContractEvaluationContext {
}

interface IContractConstraint {
    + IsSatisfied()
    + GetDescription()
}

class ConstraintFactory {
    + Add/GetConstraint()
}

class GrillAllTargetsConstraint extends IContractConstraint {
}

class FGState {
    + timeSysConfig
    + contractManConfig
    + GetDefaultSave()
    + AddValidArea/Home()
    + IsValidArea/Home()
}

class FG_GM {
    + Init()
    + SpawnWristMenu()
    + InitSaveState()
    + TransitionToLevel()
    + OnSceneLoaded()
    + QUIT()
    + SetUpCallBacks()
    + RemoveCallBacks()
}

class FileIoHandler {
    + Save/LoadFGState()
    + Save/LoadHomeVaultFile()
```

```
    + Save/LoadPlayerQuickbelt()
}

class GamePlannerLite

' Relationships

FG_GM *-- SceneManip
FG_GM *-- MerchantSystem
FG_GM *-- MoneySystem
FG_GM *-- ReputationSystem
FG_GM *-- FGTimeSystem
FG_GM *-- FGContractManager
FG_GM *-- FGState
FG_GM *-- FileIoHandler

WristUi --> FG_GM
WristUi --> FGTimeSystem
WristUi --> MoneySystem
WristUi --> FGContractManager

SceneManip --> GamePlannerLite
SceneManip --> MerchantSystem

FGContractManager --> FGContractTemplateFactory
FGContractManager --> ReputationSystem

FGContractTemplateFactory --> FGContractTemplate

FGContractManager --> ContractEvaluationContext
FGContractManager --> ConstraintFactory
IContractConstraint --> ContractEvaluationContext
ConstraintFactory --> IContractConstraint

FGTimeSystem --> FGContractManager : subscribes

FileIoHandler --> FGState

@enduml
```